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A lightweight and efficient gesture recognizer for traffic police commands using spatiotemporal feature fusion

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In response to the demand for efficient and accurate recognition of traffic police gestures by driverless vehicles, this paper introduces a novel traffic police gesture recognition framework (Novel Traffic Police Gesture Recognizer, NTPGR). Initially, keypoints related to traffic police gestures are extracted using the Efficient Progressive Feature Fusion Network (EPFFNet), followed by feature modeling and fusion to enable the recognition network to better learn the temporal characteristics of gestures. Additionally, a convolution network branch and a hybrid attention branch are incorporated to further extract skeleton information from the traffic police gesture data, assign different temporal weights to key frames, and enhance the focus on important channels. Finally, in conjunction with Long Short Term Memory (LSTM), a multi-branch gesture recognition network, termed the Multi-Sequence Gesture Recognition Network (MSNet), is proposed to facilitate the integration of three branches of gesture features, thereby enhancing the targeted extraction of temporal characteristics in traffic police gestures. Experimental results indicate that NTPGR achieves 97.56% and 96.76% accuracy on the Police Gesture Dataset and UTD-MHAD Dataset, respectively, as well as average response times of 0.76s and 0.74s. It not only recognizes traffic police gestures in real-time with high efficiency but also demonstrates strong robustness and Credibility in recognizing gestures in complex environments and dynamic scenarios.

Keywords Gesture Recognition, Traffic Police Command Gesture, Deep learning, CNN, Classification

Gestures are an inherent way for humans to communicate and interact through movements of fingers, palms, arms, heads, facial expressions, or even the entire body[1]. They are particularly important in human interaction, especially when language is restricted or impractical. The meaning conveyed by gestures is often context-dependent, and factors like individual habits and cultural variations add layers of complexity, making gesture recognition a challenging task in computer vision and artificial intelligence. As autonomous driving technology progresses, the safety and reliability of self-driving vehicles increasingly depend on their ability to accurately detect and respond to external cues, such as traffic police gestures. These gestures, while crucial in guiding traffic, present a complex structure and are highly sensitive to external conditions such as lighting, weather, and background noise. As a result, the accurate recognition of traffic police gestures has become a key metric in evaluating the effectiveness of autonomous driving systems.

Current approaches for recognizing traffic police gestures can be categorized into two main types: sensor-based methods and deep learning-based methods. Sensor-based techniques rely on inertial sensors to capture the movements of the traffic police's arms and are further divided into wearable sensor methods and depth sensor methods. The wearable sensor approach typically measures changes in the acceleration and angular velocity of the arms and torso, utilizing devices such as accelerometers, state machines, and digital compasses for gesture recognition[2,3]. The depth sensor approach uses sensor data and computer models of 3D scenes or coordinates to represent the positions of body parts for gesture recognition. Techniques include locating body parts through depth histograms, Bayesian skin color detection, and applying methods like extreme learning machines for recognizing traffic police gestures[4], as well as utilizing Kinect for this purpose[5,6]. While these methods generally offer high speed and accuracy in recognition, their broader adoption is constrained by the high hardware costs and the physical demands placed on users. The second approach is gesture recognition based on deep learning. With deep learning achieving significant success in areas like image processing and natural language processing,

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researchers have begun exploring its application in gesture and action recognition[7]. Among current deep learning methods, some focus solely on the spatial features of the human body and gestures for classification and recognition[8–11]. Others incorporate both spatial features and temporal motion information across frames to classify and recognize gestures[12–15,17,17,18]. While the aforementioned methods have achieved some success in recognizing traffic police gestures, their performance enhancements typically focus on individual aspects, such as optimizing model architecture or improving feature extraction. This narrow approach often neglects the influence of complex environmental factors, including lighting variations, background interference, and subtle gesture differences, which can hinder the model's generalization capabilities across diverse settings. Additionally, these methods tend to exhibit high computational complexity and low real-time performance, failing to meet the response speed requirements for real-time applications. Consequently, achieving a balance between accuracy and real-time performance remains a pressing challenge in the field of traffic police gesture recognition.

Morever, in human pose estimation and gesture recognition tasks, various feature fusion methods have been widely applied to enhance the integration of multi-scale features. For instance, ADSFNet[19] captures differences between bitemporal features through the specific feature enhancer (SFE) and integrates differential features across various levels using the multilevel differential feature adaptive fusion module (MDFAFM), thus improving the accuracy of change detection. MSFF-Net[20] constructs a Bi-directional Temporal-Spatial Feature Pyramid (BiTSFP) for fusing shallow spatial information with deep semantic information, while introducing an Attention-Guided Fusion (AGF) mechanism to adaptively learn the fusion weights of adjacent features to integrate them appropriately. Furthermore, the Frame-wise Attention (FA) module further enhances the utilization efficiency of multi-scale temporal features, improving the expressive power of temporal information. SS-Net[21] integrates deep learning-based saliency and JND (SJAQP), where Adaptive Quantization Parameter (QP) selection algorithm is used to dynamically adjust quantization parameters, and the perceptual distortion terms is computed with JND during Rate-Distortion Optimization (SJRDO), optimizing rate–distortion balance to achieve efficient perceptual video compression. ALPD-Net[22] combines asymmetric light-aware interaction (ALI) module with Channel-Space Feature Fusion module (CSFF), where ALI reduces interference between the RGB and thermal imaging modalities in an asymmetric manner, and CSFF selects and fuses information from different modalities in both channel and spatial dimensions, thus enhancing RGB-T salient object detection performance. CFANet[23] employs a novel cross-layer feature aggregation (CFA) module for feature fusion, aggregating features of different scales by reducing semantic gaps, thus replacing conventional feature fusion methods and achieving more precise object detection. VSS-Net[24] uses a Context-Semantics Interaction Layer (CSIL) for feature fusion, achieving sophisticated information exchange across frame-level temporal cues and video-level semantic representation, guaranteeing informative representations and boosting the sensitivity to important segments. Additionally, FFAGRNet[25] combines full-scale feature aggregation (FFA) with grouping feature reconstruction (GFR) module, enhancing object detection quality through cross-layer feature adaptation and fine-grained feature recombination.

To address the aforementioned issues and drawing on existing feature fusion methods, we present a lightweight and efficient gesture recognition method for computer vision, referred to as the Novel Traffic Police Gesture Recognizer (NTPGR). The recognizer comprises four key steps: gesture data preprocessing, gesture spatial feature extraction, spatial feature modeling and fusion, and gesture time sequence feature extraction. During the data preprocessing stage, image processing is applied to the gesture data to filter out key frames. Keypoints of the human body are then extracted. Next, spatial features are modeled and fused before being input into the proposed gesture recognition network for temporal features extraction to classify traffic police gestures. This effective combination of steps enables the recognition system to achieve optimal performance in the traffic police gesture recognition task. The main results and contributions of this paper are as follows:

- Building on the SeriesNet network[26], we propose a convolution network branch specifically for traffic police gesture recognition to effectively capture the spatial context of human postures within gesture data, significantly enhancing recognition performance. Additionally, we design a hybrid attention branch that incorporates both temporal and channel attention. This approach allows the model to assign varying attention weights to different key frames while also emphasizing important channels in each frame, thereby improving the model's ability to identify critical gesture features. Finally, we propose a new composite network structure consisting of two sub-networks: the Efficient Progressive Feature Fusion Network (EPFFNet)[27] and the Multi-Sequence Gesture Recognition Network (MSNet). The EPFFNet is tasked with extracting the skeleton data of the traffic police, while the MSNet comprises three branches. This setup enables further extraction of temporal features from the skeleton data obtained through the EPFFNet, facilitating effective fusion of multi-branch gesture features.
- The NTPGR achieves recognition accuracies of 97.56% and 96.76% on the the Police Gesture Dataset[15] and UTD-MHAD dataset[28], respectively, with average response times of just 0.76 s and 0.74 s. Compared to other gesture recognition networks, NTPGR demonstrates exceptional recognition performance and real-time capabilities, fully validating the algorithm's balance between performance and efficiency.

Methods

Overview of NTPGR framework

In this study, We use the publicly available Police Gesture Dataset and the UTD-MHAD dataset. Additionally, to enhance the performance of traffic police gesture recognition detection, we develop an efficient traffic police gesture recognition system called the Novel Traffic Police Gesture Recognizer (NTPGR). This system primarily consists of four components: gesture data preprocessing, gesture spatial feature extraction, spatial feature modeling and fusion, and gesture time sequence feature extraction. The overall process of NTPGR is illustrated in Figure.1.

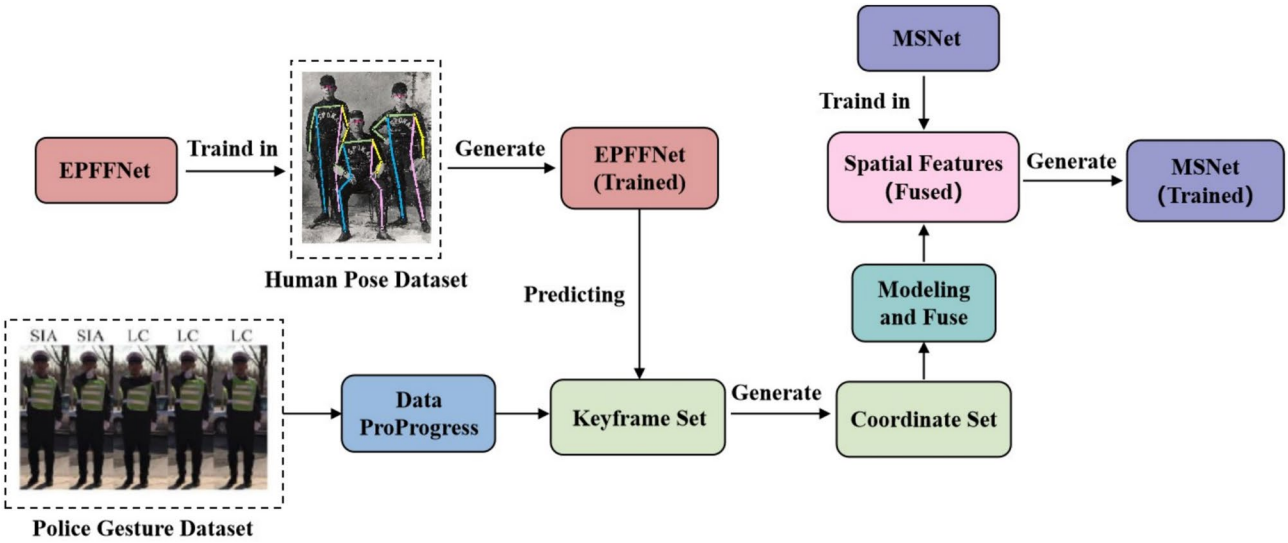


Fig. 1. Illustrating the architecture of the proposed NTPGR. It encompasses four key steps: gesture data preprocessing, gesture spatial feature extraction, spatial feature modeling and fusion, and gesture time sequence feature extraction.

We first use a data preprocessing module to select the key frames of all gestures, reducing redundant information while retaining the key dynamic changes of the actions. Then, we employ the EPFFNet network to extract the keypoint coordinates of the human body, thereby obtaining the skeletal spatial features of the gestures. During the feature modeling and fusion stage, we model the spatial features of the human skeleton by calculating the angle and distance features between keypoints, capturing the spatial relationships of gesture shapes. Unlike traditional methods that simply concatenate or weight-sum features, we adopt a fusion strategy based on semantic consistency, allowing different scales of spatial information to complement each other and effectively enhance the model's spatial representation capability. Next, the fused spatial features are fed into the MSNet network for temporal feature extraction. MSNet employs a three-branch architecture, including a convolution network branch, a sequence network branch, and a hybrid attention branch. The convolution network branch further extracts local spatial features, enhancing the expression of gesture shapes, while the sequence network branch captures dynamic changes in the temporal dimension, modeling the temporal dependencies of gestures. However, traditional spatiotemporal fusion methods typically use simple temporal concatenation or weighted accumulation, which may lead to the loss of critical information or the introduction of redundant information. In contrast, MSNet employs collaborative learning through its three-branch structure, effectively modeling the spatiotemporal relationships at both local and global scales, preventing information loss and improving the model's ability to distinguish complex gestures. Additionally, the hybrid attention branch consists of temporal attention and channel attention modules. The temporal attention module adaptively adjusts the weights of different time frames using Softmax and fully connected layers, enabling the model to focus on key time segments while ignoring irrelevant or noisy frames. The channel attention module learns the importance of different feature channels to enhance the effective representation of gesture features. This attention mechanism not only improves the semantic consistency between deep and shallow features but also enhances the model's robustness to complex environmental factors such as occlusion and lighting changes.

Therefore, NTPGR extracts spatial features using EPFFNet, enhances feature representation through the modeling method, and employs MSNet for temporal modeling, ultimately improving the model's dynamic perception capability through a hybrid attention mechanism. This method not only addresses the issue of insufficient spatiotemporal feature fusion in traditional approaches but also demonstrates stronger stability and accuracy in complex environments, as detailed in the subsequent chapters. This process is outlined in Algorithm 1.

Input: Skeleton dataset $D_{skeleton}$, Traffic gesture dataset $D_{gesture}$,
Output: Trained MSNet Θ_{MSNet}, Gesture recognition results $R_{gesture}$

$\Theta_{\text{EPFF}} \leftarrow \text{Train}(\mathscr{D}_{skeleton}, \text{EPFFNet})$
$K_{gesture} \leftarrow \text{Preprocess}(\mathscr{D}_{gesture})$
$P_k \leftarrow \text{Train}(\Theta_{\text{EPFF}}, K_g)$
$F_s \leftarrow \text{Model}(P_k)$
$\Theta_{\text{MSNet}} \leftarrow \text{Train}(F_s, \text{MSNet})$
$R_{gesture} \leftarrow \text{Predict}(\Theta_{\text{MSNet}}, \mathscr{D}_{gesture})$
return Θ_{MSNet}, $R_{gesture}$

Algorithm 1. NTPGR

Gesture data proprocessing

In dynamic gesture sequences, repetition or similar issues arise in consecutive frames, resulting in redundant data and increasing the computational burden on the network[18]. To solve this issue, we employ the frame difference method to extract key frames from the gesture sequence, replacing the original video sequence with the key frame sequence. This approach effectively removes redundant information and significantly enhances the efficiency of the network training process. we describe this process in Algorithm 2.

Input: Video sequence $V = \{F_1, F_2, \ldots, F_n\}$, Threshold function $\tau(\mu, \sigma)$, frame index $i = 0$
Output: Key Frames K

Initialize key frame set $K = \emptyset$
for $i = 1$ to $n - 1$ **do**
  $G_i \leftarrow \text{Grayscale}(F_i)$
  $G_{i+1} \leftarrow \text{Grayscale}(F_{i+1})$
  $B_i \leftarrow \text{GaussianBlur}(G_i)$
  $B_{i+1} \leftarrow \text{GaussianBlur}(G_{i+1})$
  $\Delta_i = B_{i+1} - B_i$
  $\mu_i = \frac{1}{|B_i|} \sum_{p \in B_i} \Delta_i(p)$
  $\sigma_i = \sqrt{\frac{1}{|B_i|} \sum_{p \in B_i} (\Delta_i(p) - \mu_i)^2}$
  $\lambda_i = \tau(\mu_i, \sigma_i)$
  if $\Delta_i > \lambda_i$ **then**
    Add F_{i+1} to K
  end if
end for
return K

Algorithm 2. Extract Key Frames from Video Sequence

This algorithm condenses all frames in the gesture video sequence into key frames, filtering out input feature data that is more beneficial for the traffic police gesture recognition task and enhancing the model's performance. First, all frames in the gesture video sequence are read, and image preprocessing is applied, including steps such as grayscale processing and Gaussian filtering. Next, the threshold λ is determined by calculating the pixel difference between each pair of adjacent frames and then computing the mean and standard deviation of all pixel differences, which can be expressed as $\lambda = 0.75\sigma + \mu$. σ denotes the standard deviation, and μ indicates the mean value. we compare the pixel difference of each frame pair against the threshold. If the difference exceeds the threshold, the current frame is saved as a key frame; otherwise, the process continues by comparing the pixel difference of the next frame pair with the threshold. This iteration continues until all frames have been assessed for pixel differences. If any frames remain unexamined, the algorithm returns to the pixel difference calculation step and repeats the threshold comparison. Once all comparisons are completed, the identified key frames are output for subsequent processes.

Gesture spatial feature extraction

To achieve a balance between recognition performance and model complexity, and to support deployment in resource-constrained environments, we employs the Efficient Progressive Feature Fusion Network (EPFFNet) as the skeletal keypoint extraction module in the proposed NTPGR framework. EPFFNet is a lightweight multi-branch architecture characterized by low computational cost and parameter count. It maintains high processing efficiency while enabling precise localization of human keypoints. Demonstrating strong performance on standard human pose estimation benchmarks, EPFFNet provides a solid foundation for skeleton-based modeling in traffic police gesture recognition.

Another advantage of EPFFNet lies in its multi-scale feature fusion capability. While preserving high-resolution feature streams, the network employs a ogressive feature fusion mechanism to effectively integrate semantic information from different scales. This design enhances the model's sensitivity to local details under complex conditions and improves its robustness against gesture blurring and pose occlusion. In addition, EPFFNet incorporates Mixed Efficient Channel Attention(MECA) and Efficient Coordinate Attention(ECA) that guides the network to focus on regions more critical for gesture recognition, thereby further strengthening feature extraction.

To address the dual demands of real-time performance and environmental adaptability in traffic police gesture recognition, EPFFNet is designed to balance accuracy with inference speed. Its low latency and efficient resource

utilization make it suitable not only for deployment on conventional desktop platforms but also on resource-constrained edge devices. After data preprocessing, the extracted keyframe sequences are fed into EPFFNet for skeletal keypoint localization, providing accurate and stable structural features to support the subsequent temporal modeling module. The network structure of EPFFNet is illustrated in Figure.2.

Spatial feature modeling and fusion

Dynamic gestures consist of two primary components: space and time. From a spatial viewpoint, different gesture movements result in changes to the angles between adjacent bones and the distances between joint points. Consequently, angle features and distance features can effectively describe the spatial relationships of dynamic gestures. From a temporal perspective, variations in gesture sequences lead to different timing trends for each keypoint. Thus, combining spatial features with temporal features can enhance the accuracy of traffic police gesture recognition.

Distance feature

To comprehensively describe the spatial relationships of traffic police gestures, we model distance features by calculating seven sets of distances related to gesture action features. The hinge dependencies between each skeleton and their corresponding distance features are presented in Table 1.

Using the coordinate information of each keypoint extracted by EPFFNet, the Euclidean distance between adjacent keypoints can be calculated, as illustrated in Equation (1).

$$d = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{1}$$

In this equation, d represents the Euclidean distance, while (x_i, y_i) and (x_j, y_j) denote the coordinates of two different human body keypoints.

Thus, the distance features calculated for each gesture key frame are presented in Equation (2), where Distance denotes the distance feature set and d_n represents the distance between each keypoint.

$$Distance = (d_1, d_2 ..., d_n) \tag{2}$$

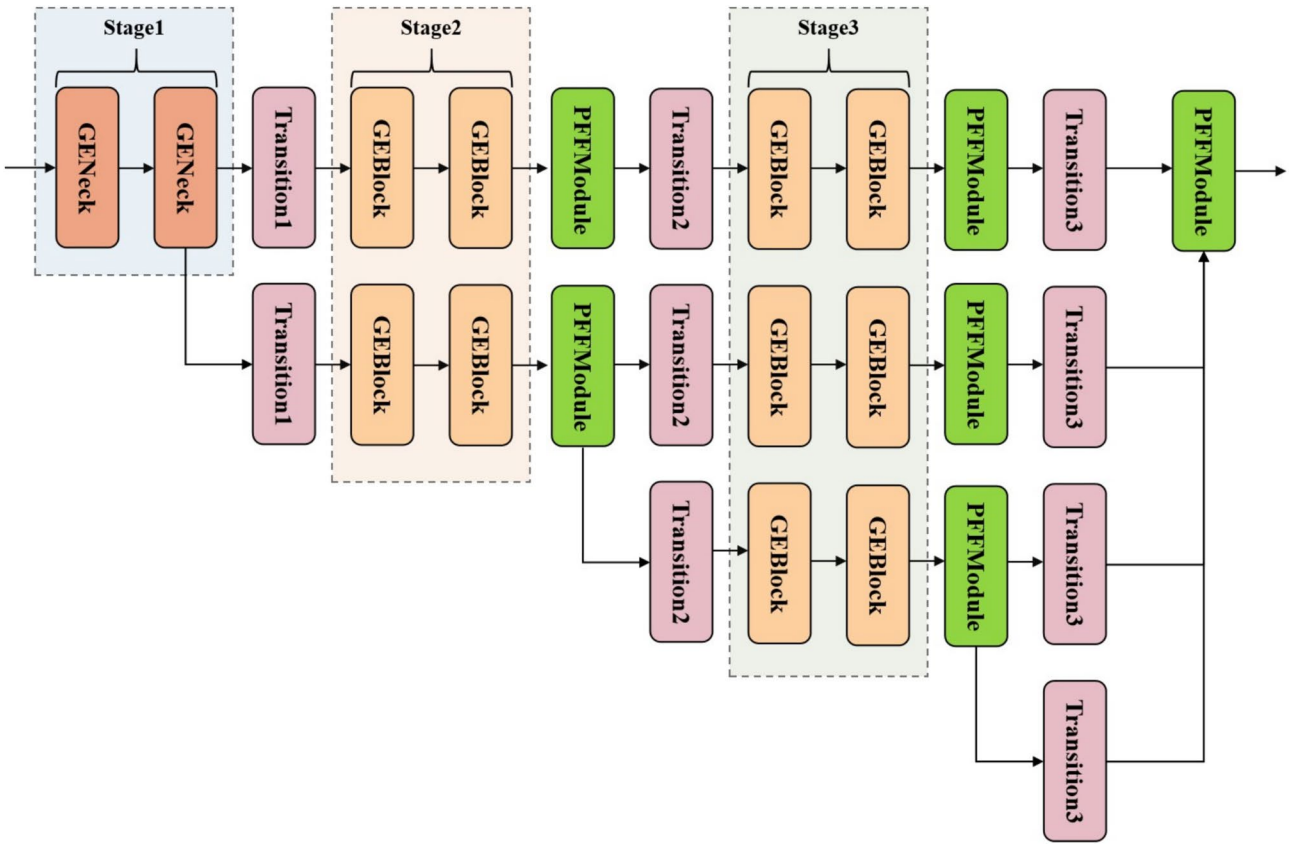


Fig. 2. Illustrating the architecture of the proposed EPFFNet. It consists of parallel high-to-low resolution subnetworks with repeated information exchange across multi-resolution subnetworks(multi-scale fusion). The horizontal and vertical directions correspond to the depth of the network and the scale of the feature maps, respectively.

Keypoints Pair	Keypoints1	Keypoints2	Distance
4–9	Right Wrist	Right Hip	Right Hand And Wrist
7–12	Left Wrist	Left Hip	Left Hand And Wrist
2–4	Right Shoulder	Right Wrist	Right Hand And Shoulder
5–7	Left Shoulder	Left Wrist	Left Hand And Shoulder
0–4	Nose	Right Wrist	Head And Right Hand
0–7	Nose	Left Wrist	Head And Left Hand
4–7	Right Wrist	Left Wrist	Left Hand And Right Hand

Table 1. Hinge dependency table between skeletons. The distance features of the skeleton are calculated using hinge dependencies.

Angle feature

Given that the direction of gravitational acceleration is always perpendicular to the ground, we determine the angular characteristics of each skeleton by calculating the angle between the skeleton and gravitational acceleration[29]. The calculation formula is provided in Equation (3).

$$F_{2,i} = \{\frac{S_i \cdot d}{|S_i||d|}, \frac{S_i \times d}{|S_i||d|}\}, F_{2,i} \in F_2 \tag{3}$$

To maintain feature continuity, we describe the angular characteristics between the skeletons by calculating the values of trigonometric functions related to the skeleton and gravitational acceleration. Here, S_i denotes the skeletal vector of the i-th individual, d denotes the unit vector in the same direction as gravitational acceleration, while $\frac{S_i \cdot d}{|S_i||d|}$ and $\frac{S_i \times d}{|S_i||d|}$ represent the cosine and sine values of the angle between each skeleton vector and the direction of gravitational acceleration, respectively. $F_{2,i}$ denotes the concatenated vector of angular features for each skeleton relative to gravity, and F_2 represents the set of all skeletal direction vectors.

Spatial context feature fusion

we employ a feature coding method to effectively integrate the extracted angular and spatial features, primarily utilizing the Fisher Vector (FV)[30]. The FV algorithm not only captures the frequency of each codebook's occurrence but also retains the distribution and occupancy frequency of features within the codebook, representing the high-dimensional relationships between features and codebooks. This capability makes FV particularly well-suited for gesture recognition tasks, especially for video gesture classification problems, as it can characterize richer and more diverse video features by capturing subtle changes and temporal dynamics between frames. Additionally, the FV algorithm exhibits strong robustness and generalization capabilities, contributing to improved recognition accuracy and real-time performance of the model. Therefore, we select the FV algorithm to encode and fuse the extracted angle and distance features. The specific calculation formula is presented in Equation (4).

$$F = \Phi\left(F_1 + F_2\right) \tag{4}$$

In this context, F denotes the fused spatial context feature, while F_1 and F_2 represent the angle feature and distance feature, respectively. Φ indicates the FV algorithm.

Gesture time sequence feature extraction

Since the data used in this study primarily consists of video sequence data, it is necessary to design a model that effectively ensures the continuity of the sequence data. Therefore, we proposes a Multi-Sequence Gesture Recognition Network (MSNet), specifically designed for the temporal feature extraction of traffic police gestures. This network fully leverages the spatiotemporal features of the data, incorporating both the continuity features of the sequence and the spatial features within the sequence, thereby improving the overall recognition accuracy. The network structure is illustrated in Figure.3.

MSNet models the fused spatial context features to efficiently capture the dynamic changes of traffic police gestures in the temporal domain, thereby significantly improving the stability and robustness of the recognition. Specifically, the network consists of the following three branches:

- **Convolution Network Branch:** This branch consists of multiple convolution module layers, designed to further learn the local and global spatial features of time series data and improve the network's generalization capability in gesture recognition tasks.
- **Sequence Network Branch:** This branch primarily consists of LSTM. It employs LSTM to learn the temporal characteristics of gesture sequences and capture the temporal dependencies between frames, aiding in understanding the motion trajectory and dynamic variations of gestures.
- **Hybrid Attention Branch:** This branch assigns varying attention weights to key frames and enhances the focus of each key frame on important channels, thereby improving the network's recognition performance.

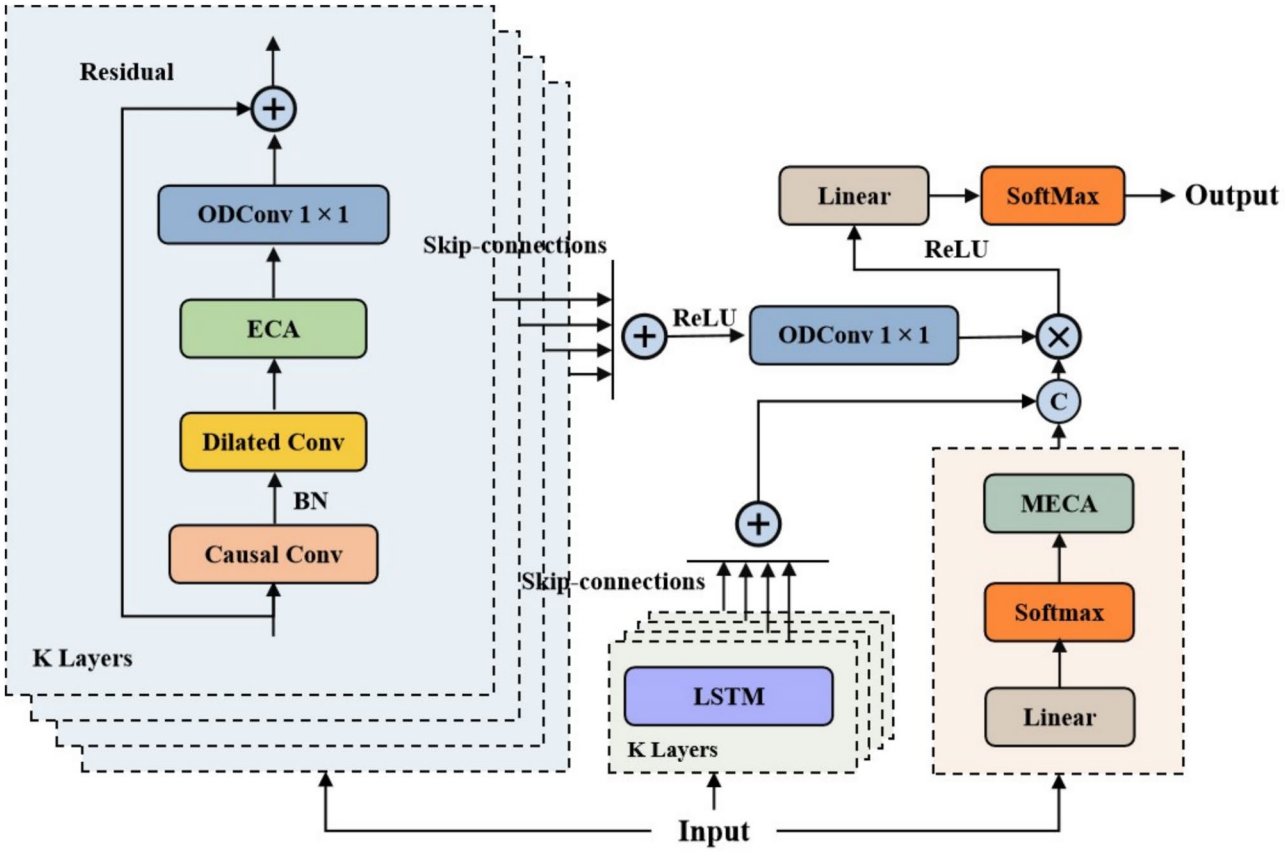


Fig. 3. Illustrating the architecture of the proposed MSNet. It comprises three sub-networks: the convolution network branch, the sequence network branch, and the hybrid attention branch.

Initially, data with varying temporal characteristics are fed into the convolution network branch to enhance the understanding of the spatial characteristics of each frame within the gesture sequence data. Next, the data is directed to the sequence network branch, enabling the model to capture the dependencies between different frames and extract temporal features for gesture recognition. Additionally, the fused spatial data is input into the hybrid attention branch to provide the network with diverse attention capabilities regarding channel features and temporal features, further improving the network's effectiveness in learning spatiotemporal features. Subsequently, the features produced by the convolution network branch undergo processing through a dynamic convolution (Omni-dimensional Dynamic Convolution, ODConv)[31] to extract high-level spatial context features and compress the features. Finally, the processed features are matrix-multiplied with the output from the other two branches and passed through a ReLU layer to enhance the nonlinear fitting capability of the recognition network. The resulting data is then processed through a linear layer and a SoftMax layer to determine and output the final gesture recognition category. Additionally, the input dimensions for each branch of MSNet are [32, 64, 27], where 32 represents the batch size, 64 denotes the number of feature channels, and 27 corresponds to the number of sequence frames, i.e., the number of features fused from the coordinate sets of 27 action frames. The final output dimensions of MSNet are [32, 8], indicating the prediction probabilities for 8 gesture classes.

The three branches of the MSNet network effectively combine the spatial context features and temporal continuity features of gesture sequence data. Using the fused features, we compute the predicted probability for each type of traffic police gesture as outlined in Equation (5).

$$o_r^{all} = s\left(o_1 + o_t + o_3\right) \tag{5}$$

Here, s denotes the softmax function, and o_r^{all} signifies the probability that the current gesture corresponds to each gesture class. Finally, we identify the gesture with the highest predicted probability as the final predicted gesture according to Equation (6).

$$o_r^{max} = \begin{cases} argmax\left(o_r^{all}\right), & max\left(o_r^{all}\right) > \delta \\ 0, & max\left(o_r^{all}\right) < \delta \end{cases} \tag{6}$$

In this context, o_r^{max} represents the final gesture classification output, δ indicates the action confidence threshold, and a value of zero denotes the standby gesture.

Convolution network branch

While the SeriesNet network combines the strengths of convolution and temporal networks, demonstrating strong detection performance in gesture recognition tasks, it has limitations in extracting temporal and spatial features, particularly in complex dynamic scenes. To address this issue, we optimize the convolution network branch of SeriesNet and propose an enhanced branch specifically tailored for traffic police gesture recognition. This new branch consists of multiple convolution module layers, enabling deeper exploration of the spatial context features of human postures in traffic police gesture data. This significantly improves the extraction of spatial features and enhances the overall detection accuracy and stability of the model. The specific structure of the convolution module layers is shown in Figure. 4(b).

The convolution module layer that comprises the convolution network branch consists of two main components: the trunk and the residual connection. Initially, the input data passes through a causal convolution layer to ensure that the gesture sequence data maintains causal relationships, enabling the network to learn features across various time intervals. Following this, batch normalization is applied to expedite network training while enhancing the robustness and generalization capabilities of the model. Additionally, dilated convolution is employed to expand the receptive field of the network, facilitating the capture of broader spatial contextual features. Next, efficient coordinate attention is integrated to improve the ability of each convolution module layer to focus on critical information across different feature regions. Finally, dynamic convolution is utilized, allowing the convolution kernel to adapt based on the shape and size of the input features, thereby enabling more precise learning of essential information in the input data, which is particularly beneficial for the task.

Furthermore, the incorporation of residual connections is intended to mitigate issues related to network degradation. By stacking multiple convolution module layers, convolution network branch achieves an optimal receptive field size and enhances the model's capability to focus on various regional features. The calculation for the convolution module layer is represented in Equation (7), while the calculation for the output features of the convolution network branch is detailed in Equation (8).

$$M_{out} = ODConv\left(ReLU\left(ECA\left(DConv\left(BN\left(F_{in}\right)\right)\right)\right)\right) \tag{7}$$

$$o_1 = \oplus_{i=1}^{i=k} M_{out} \tag{8}$$

Here, *DConv* denotes dilated convolution, F_{in} represents the input data, M_{out} signifies the output data processed by a single convolution module layer, $\oplus$ indicates the vector splicing operation, and o_1 represents the output data obtained after the splicing process.

Sequence network branch

Dynamic traffic police gestures comprise a series of images arranged in a temporal sequence. Consequently, we design a sequence network branch based on Long Short-Term Memory (LSTM) units. This branch facilitates layer-by-layer information transfer in the time dimension by stacking multiple LSTM units, effectively retaining the characteristics of each time step. Ultimately, the outputs from each LSTM layer are integrated through splicing operations to create a more comprehensive temporal feature representation. The specific structure of the LSTM unit is illustrated in Figure. 4(a).

LSTM is a type of recurrent neural network that effectively preserves long-term memory, making it widely applicable for extracting time-related features. It retains memory content using the following formula:

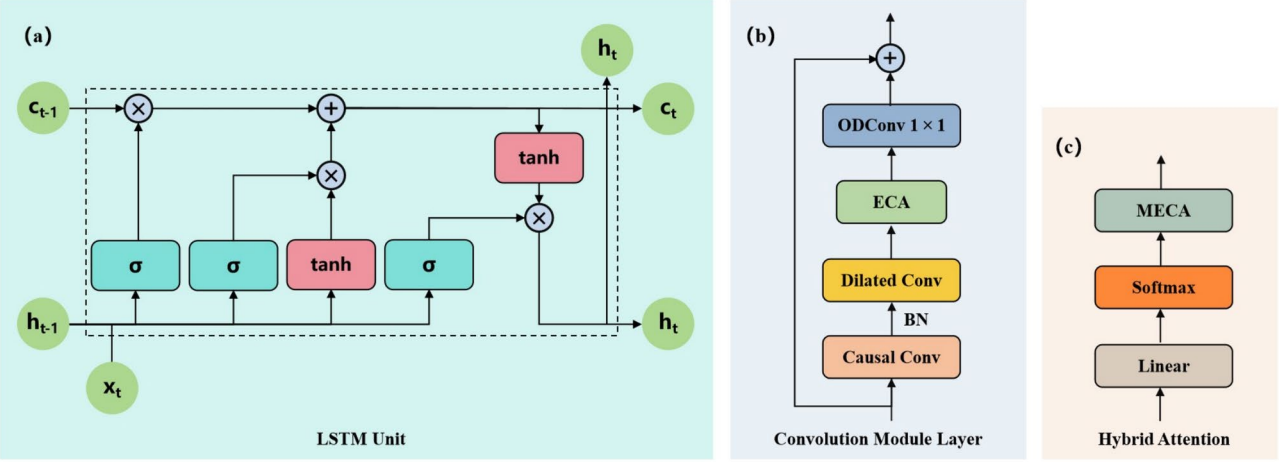


Fig. 4. The various components of the MSNet architecture. (**a**) the LSTM unit, which forms the sequential network branch. (**b**) the convolution module layer, which constitutes the convolution network branch. (**c**) the hybrid attention mechanism.

$$\begin{cases} f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) * C_{t-1} \\ i_t = \sigma\left(W_i \cdot [h_{t-1}, x_i] + b_i\right) \\ \widetilde{C}_t = tanh\left(W_c \cdot [h_{t-1}, x_t] + b_c\right) \\ C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C}_t \end{cases} \tag{9}$$

Here, W represents the weight parameter, b indicates the bias, x_i denotes the spatial feature of the traffic police gesture at time t, and h_{t-1} is the information from the previous time step (t-1), The σ symbolizes the sigmoid function, while f_t is the forget gate, i_t is the input gate, and C_{t-1} refers to the previous cell state, with $\widetilde{C}_t$ being the cell state at time t. Additionally, LSTM calculates the output gate o_t according to Equation (10).

$$o_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right) * tanh\left(C_t\right) \tag{10}$$

Among these functions, *tanh* refers to the hyperbolic tangent function, represents matrix multiplication, and $*$ signifies dot multiplication.

Hybrid network branch

In dynamic traffic police gesture recognition, temporal information is rich, and each frame of the image carries different important information through its channels. Therefore, we designs a hybrid channel attention branch aimed at enhancing the network's ability to focus on key channel information. This branch strengthens the network's attention to specific key frames and certain channel information by assigning different weights to key frames, thereby more effectively learning critical information from dynamic traffic police gestures and improving model performance. The structure of the hybrid attention mechanism is illustrated in Figure. 4(c).

The hybrid attention branch consists of two main components: the temporal attention mechanism and MECA attention[27]. The temporal attention mechanism calculates the importance scores of each time frame through the Softmax function and fully connected layers, normalizing them into weights that allow the model to automatically focus on critical moments. MECA attention, based on the output of the temporal attention, further enhances the network's learning ability for important channel features in specific key frames.

In the traffic police gesture recognition task, the importance of different frames over time changes dynamically. For instance, the starting or ending frames of a gesture typically carry more critical information, while the middle frames may be less crucial for gesture recognition in certain cases. Therefore, the network needs to dynamically adjust the attention weights of different frames based on the features of the current input data. The temporal attention mechanism in this study achieves this goal by introducing a dynamic weighting mechanism. Specifically, temporal attention first calculates the feature representation of each frame and then generates a weight vector using the Softmax function, reflecting the importance of each key frame. This allows the network to automatically adjust the weights of different frames based on the characteristics of the current scene (such as the start, continuation, or end phase of the gesture), enabling the network to focus more on key frames in the time sequence during dynamic recognition while ignoring less important frames.

In addition to the dynamic adjustment along the temporal dimension, the contribution of different channel information in traffic police gestures to the recognition result is also inconsistent. In some cases, specific image channels (such as those related to hand position or posture) may be more critical than others. Therefore, the network needs to dynamically adjust the attention weights of different channels based on the varying content of the input image. To address this, we proposes a dynamic channel attention adjustment strategy based on the MECA attention mechanism. This strategy analyzes the output of key feature channels in each frame and combines the results of temporal attention to assign a dynamic weight to each channel. In the key feature channels of each frame, important information is given higher weight, helping the network better focus on the content of these channels, thereby improving the accuracy of gesture recognition.

In this way, the hybrid attention branch can adaptively adjust the attention weights along both the temporal and channel dimensions based on the features of different gestures and changes in the current scene. This not only enhances the flexibility of the model but also enables it to better handle the dynamic variations in traffic police gesture recognition. The computation of the hybrid attention branch is presented in Equation (11), where x denotes the input data.

$$o_3 = MECA\left(Softmax\left(Linear\left(x\right)\right)\right) \tag{11}$$

Experiments and result analysis

Experimental Settings

The proposed model is implemented using Python 3.9 runs on Ubuntu 21.1, with an Intel Xeon E5 processor and dual NVIDIA GTX TITAN XP graphics cards. We conduct experiments using the PyTorch framework. The spatial feature weights for the skeleton are derived from the trained EPFFNet network, followed by training the MSNet network to obtain the temporal feature weights for the skeleton. In the experiment, the batch size is set to 8, the initial learning rate is established at 1e-3, the training period last for 210 epochs, and the dropout rate is set at 0.5.

We conduct experiments using the Police Gesture Dataset and the UTD-MHAD dataset. The Police Gesture Dataset comprises 21 short videos with a total duration of approximately 2 hours. This dataset includes 11 training sets and 10 validation sets, featuring traffic police gesture videos recorded in various complex scenes, such as those involving passing vehicles and pedestrians. Multiple subjects with diverse body types were selected for the recordings, all wearing traffic police uniforms. The dataset encompasses 8 gesture actions: Lane Change (LC), Move Straight (MS), Pull Over (PO), Slow Down (SD), Stop (S), Turn Left (TL), Turn Left to Stay (TLTS),

and Turn Right (TR), resulting in a total of 448 video sequences. This dataset is currently the most representative traffic police gesture dataset aligned with China's traffic laws.

The dynamic dataset UTD-MHAD consists of 27 action sequences performed by 8 different subjects, with each action sequence repeated 4 times, resulting in a total of 861 RGB video sequences. The data collection process covers a variety of complex backgrounds, such as blackboards, chairs, trees, and pedestrians, to increase the diversity of the scenes. Additionally, the dataset includes 16 hand gesture actions, which are: Wave, Clap, Throw, Arm Cross, DrawX, Draw Circle (counterclockwise), Boxing, Basketball Swing, Tennis Swing, Arm Curl, Push, Catch, Knock, Draw Triangle, Swipe Left, and Tennis Serve.

This dataset utilizes Edit Distance and Average Response Time (ART) as evaluation criteria. Edit Distance is commonly used to assess the differences between two sequences. By applying a series of changes to sequence X, such as insertions, deletions, and substitutions, it can be transformed to match sequence Y. The formula for calculating the accuracy of Edit Distance is presented in Equation (12).

$$AP = \frac{(N - I - D - S)}{N} \tag{12}$$

Here, N denotes the total number of poses in the video, I represents the total number of inserted poses, D indicates the total number of deleted poses, and S signifies the total number of replaced poses.

The Average Response Time (ART) measures the system's efficiency in processing a series of requests or events, representing the average duration between receiving and completing a request. The formula is presented in Equation (13).

$$ART = \sum_{i=1}^{n} T_i \tag{13}$$

In this formula, T represents the response time of each request, i denotes the request sequence number, and n refers to the total number of requests.

Analysis of experimental results

To verify the effectiveness of our proposed algorithm, we present experimental validation and visual analysis. We first use the pre-trained EPFFNet network to make predictions on the UTD-MHAD dataset and the Police Gesture Dataset to obtain the human skeleton data. To ensure the continuity of gesture actions in the time dimension, we extract a continuous sequence of 27 frames from each video as the input unit for the model. All subsequent experiments are based on this preprocessing step. Specifically, we analyze gesture recognition experiments on both the SeriesNet network and the NTPGR, recording the results for comparison to assess the recognition stability of NTPGR in various complex scenarios. Next, NTPGR is compared and analyzed against existing traffic police gesture recognition algorithms to demonstrate its effectiveness. Additionally, we conduct ablation experiments on NTPGR to validate its effectiveness in the traffic police gesture recognition task by systematically removing each module and calculating the resulting accuracy. Finally, a visual analysis of the traffic police gestures recognized by NTPGR is performed to assess its applicability in the gesture recognition task.

Gesture recognition experimental results

To verify the robustness of NTPGR in recognizing traffic police gestures across various complex environments, we utilize both the SeriesNet network and NTPGR to conduct experiments on the Police Gesture Dataset. The experimental results are recorded to compare the performance of the two algorithms in challenging scenarios, highlighting their differences in recognizing traffic police gestures. The comparison results of the gesture recognition experiments between the two algorithms are presented in Table 2.

Table 2 illustrates that the accuracy of the SeriesNet network on the Police Gesture Dataset is 91.13%, indicating that the network performs well in the traffic police gesture recognition task. However, in video files 008, 014, 016, and 018, the network's accuracy is significantly lower compared to other files. An analysis of the reasons revealed that these four videos are recorded in a specialized environment. Specific environmental factors, such as lighting, pedestrians, vehicles, trees, and railings, negatively impact the model's ability to recognize traffic police gestures. These factors introduce interference, leading to a decrease in accuracy. Figure.5 displays some complex scenes from these video files, with the elements affecting gesture recognition highlighted in the white box.

To address the impact of complex environments on the accuracy of traffic police gesture recognition, we enhance the SeriesNet network by proposing a new gesture recognition algorithm, NTPGR, and conducts experiments to evaluate its performance. The results are presented in Table 2. As shown in the table, NTPGR achieves an accuracy of 97.56% on the Police Gesture Dataset, which is 6.43% higher than that of the SeriesNet network. Furthermore, NTPGR demonstrates competitive gesture recognition accuracy compare to SeriesNet in standard scenes, and in the four challenging scenarios (008, 014, 016, and 018), NTPGR's accuracy improve by 10.00%, 14.63%, 17.50%, and 6.17%, respectively. These results clearly indicate that NTPGR maintains stable recognition performance across various complex environments.

Additionally, this study selects the two files, 014 and 016, which exhibits the lowest recognition accuracy, to conduct heat map experiments on both the SeriesNet network and the NTPGR network. The experimental results are displayed in Figure.6. The findings indicate that, in the complex scenes of files 014 and 016, the gesture recognition accuracy of the SeriesNet network experiences significant fluctuations due to the influence of five distinct special factors. Specifically, in file 014, the gesture recognition accuracy of the SeriesNet network ranges from 68% to 78%, while in file 016, the accuracy improves slightly, ranging from 66% to 82%. In contrast, the

Files	Mark Gesture Numbers	Correct Gestrues Numbers		Accuracy/%	
		SeriesNet	NTPGR	SeriesNet	NTPGR
002	80	77	80	96.25	100.00
004	50	49	50	98.00	100.00
008	80	70	78	87.50	97.50
010	72	67	70	93.06	95.22
012	88	84	86	95.45	97.73
014	82	68	80	82.93	97.56
016	80	62	76	77.50	95.00
018	81	72	77	88.89	95.06
102	82	80	81	97.56	98.78
104	83	80	81	96.39	97.59
Total	778	709	759	91.13	97.56

Table 2. Comparative results of gesture recognition accuracy of SeriesNet and NTPGR for each video data in the Police Gesture Dataset. All gesture data consist of video files.



Fig. 5. Special factors within the complex environment interfere with the model's gesture recognition in the Police Gesture Dataset. The white boxes highlight these factors, including lighting, pedestrians, vehicles, trees, and railings.

NTPGR network exhibits more stable performance in both files. For file 014, the gesture recognition accuracy of the NTPGR network ranges from 95% to 97%, and in file 016, it further increases, reaching between 96% and 99%. These results indicate that the five special factors present in complex scenes significantly interfere with the SeriesNet network, while the NTPGR network maintains high recognition accuracy in challenging environments, demonstrating its robustness and stability.

Comparision with the state of the art

To comprehensively evaluate the performance of the model, we not only reproduce several mainstream gesture recognition network models from recent years on the Police Gesture Dataset but also introduce a more challenging additional dataset, UTD-MHAD. This dataset includes complex scenarios such as gesture occlusion, dynamic backgrounds, and invisible gestures, which better simulate the application of traffic police gesture recognition systems in real-world environments. The experimental results are shown in Table 3. By introducing



Fig. 6. The influence of special factors on gesture recognition by SeriesNet and NTPGR. The four heat maps display the accuracy distribution of eight gesture recognitions for both networks under five special factors in videos 014 and 016.

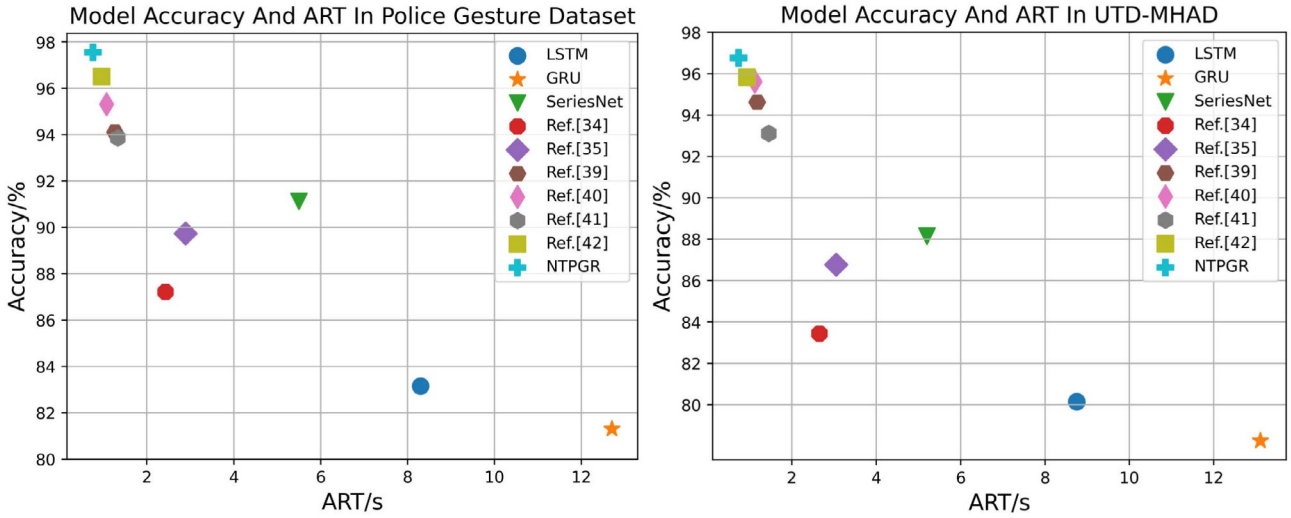
the innovative method we proposed, NTPGR achieves a better balance between recognition performance and average response time, as shown in Figure.7

Additionally, to highlight the effectiveness of the proposed method, we perform performance comparisons from two perspectives: sequence models and composite models, where the composite model refers to a model that combines sequence models with human pose estimation networks. The experimental results show that NTPGR achieves an accuracy of 97.56% in the Police Gesture Dataset, significantly improving upon the baseline model, SeriesNet. In the experiments on the UTD-MHAD dataset, despite the inclusion of complex factors such as occlusion and dynamic backgrounds, the model's accuracy declines but still maintains relatively stable recognition performance. Specifically, in the Police Gesture Dataset, NTPGR improves by 14.41%, 16.26%, and 11.92% compared to LSTM, GRU, and LSTM combined with GRU[32], respectively. On the complex UTD-MHAD dataset, NTPGR still demonstrates significant performance improvement over these three algorithms.

Compared to other skeleton-based traffic police gesture recognition models, the proposed NTPGR still demonstrates the best recognition performance on both the Police Gesture Dataset and the UTD-MHAD dataset. Specifically, compared to PKEN combined with Bidirectional LSTM[33], NTPGR improves performance on these two datasets by 6.52% and 8.22%, respectively. When compared to ResNet part localizer combined with LSTM[34] and DenseNet part localizer combined with LSTM[35], NTPGR achieves improvements of 10.34% and 7.82% on the Police Gesture Dataset, and 13.31% and 9.99% on the UTD-MHAD dataset. Furthermore, compared to End to end 3D Convolution[36], NTPGR's performance is improved by 16.40% and 19.60% on the two datasets, respectively. On the Police Gesture Dataset, NTPGR outperforms the high-performance gesture recognition composite algorithms in[15,37–40] by 6.38%, 4.12%, 15.16%, 3.44%, and 2.24%, while on the UTD-MHAD dataset,

	Method	Police Geture Dataset	UTD-MHAD
		Accuracy/%	
Sequence Models	LSTM	83.15	80.15
	GRU	81.30	78.27
	LSTM+GRU[32]	85.64	82.84
Composite Models	SeriesNet	91.13	88.16
	PKEN+Bidirectional LSTM[33]	91.04	88.54
	ResNet Part Localizer+LSTM[34]	87.22	83.45
	DenseNet Part Localizer+LSTM[35]	89.74	86.77
	End to end 3D Convolution[36]	81.16	77.16
	Hyperpose+GRU[37]	93.44	91.94
	Convolutional LSTM[38]	82.40	78.90
	PKEN+LSTM[15]	91.80	88.38
	PKEN+LSTM+GPEN[39]	94.12	94.62
	OpenPose+Angles+Trajectories[40]	95.32	95.62
	RCNN[41]	92.34	91.50
	CFF-RCNN[41]	93.87	93.12
	4-Stream Architecture[42]	96.51	95.83
	NTPGR	**97.56**	**96.76**

Table 3. Comparisons with state-of-the-art sequence models and composite models on the Police Gesture Dataset and the UTD-MHAD dataset. Where the composite model combines a sequence model with a human pose estimation model.



(a) Models in Police Gesture Dataset **(b)** Models in UTD-MHAD dataset

Fig. 7. A comparison of average response time and accuracy between mainstream gesture recognition network models. Result measured in the Police Gesture Dataset and UTD-MHAD dataset.

the improvements are 8.38%, 4.82%, 17.86%, 2.14%, and 1.14%. Finally, when compared to RCNN[41] and CFF RCNN[41], NTPGR shows improvements of 5.22% and 3.68% on the Police Gesture Dataset, and 5.26% and 3.64% on the UTD-MHAD dataset. Compared to the latest 4-Stream Architecture[42], NTPGR achieves a performance improvement of 1.05% and 0.93% on the two datasets. These experimental results indicate that while the model's performance may decrease in complex environments, NTPGR still demonstrates greater robustness compared to traditional methods. Additionally, the model effectively adapts to variations in different scenarios, especially when handling gestures with occlusion and dynamic backgrounds, where it shows higher accuracy and stability than other methods. Our future research could further enhance the model's performance under these complex conditions through data augmentation and scene-adaptive techniques.

To further validate the real-time effectiveness of the proposed algorithm in practical applications, we conduct comparative experiments between NTPGR and several recent mainstream gesture recognition models on both the Police Gesture Dataset and the UTD-MHAD dataset. All experiments are conducted under identical

Method	Police Gesture Dataset			UTD-MHAD		
	ART/s					
	GPU	CPU	Mobile Device	GPU	CPU	Mobile Device
LSTM	8.30	28.50	51.80	8.75	30.40	56.40
GRU	12.70	42.30	78.90	13.10	46.21	84.11
SeriesNet	5.50	17.40	35.80	5.20	18.10	37.41
Resnet Part Localizer+LSTM	2.43	8.20	17.30	2.65	9.10	19.14
DenseNet Part Localizer+LSTM	2.89	10.10	20.40	3.05	10.80	21.02
PKEN+LSTM+GPEN	1.26	5.40	11.80	1.18	5.53	12.00
Openpose+Angles+Trajectories	1.07	6.80	16.50	1.12	7.11	17.12
CFF-RCNN	1.33	**3.90**	**8.90**	1.45	**4.19**	**9.89**
4-Stream Architecture	0.96	7.20	18.40	0.94	6.17	19.15
NTPGR	**0.76**	4.80	10.40	**0.74**	4.91	11.32

Table 4. Comparisons of average response time with state-of-the-art models on the Police Gesture Dataset and the UTD-MHAD dataset.

Method	Components			Accuracy/%
	Data Proprocessing	EPFFNet	MSNet	
NTPGR			✓	92.60
	✓		✓	93.45
		✓	✓	95.90
	✓	✓	✓	**97.56**

Table 5. Ablation study performed by removing various key steps of NTPGR.

conditions to ensure the comparability of results. The same traffic police gesture video data was used for testing, and we compare the average response times of each model. The experimental results are shown in Table 4.

In the standard experimental environment (NVIDIA GTX TITAN XP with 24GB of memory), the average response time of NTPGR on the two datasets are 0.76 s and 0.74 s, respectively, significantly outperforming other gesture recognition models. Compared to the baseline model, SeriesNet, the response time of NTPGR is reduced by more than seven times, further validating the proposed algorithm's effectiveness in application deployment and demonstrating its superior real-time performance. Additionally, to evaluate the adaptability of NTPGR in different computational environments, we conduct further tests in a CPU environment (Intel Core i7 with 8GB of memory). The results shows that the average response time increased to 4.80 s and 4.91 s, respectively. Despite the slower computation speed, the model remains functional in resource-constrained environments. Although the response time is longer, the test demonstrates the model's adaptability on low-power devices. Finally, experiments conducted on mobile devices (Vivo S10, Dimensity 1100, 2.6GHz Cortex-A78) reveal that, due to computational limitations, the inference time further increases to 10.40 s and 11.32 s. This indicates that there is still room for optimization in deploying the model on low-power devices. To achieve real-time performance, future work may focus on optimizing the model through techniques such as quantization, pruning, reducing input resolution, or simplifying the network architecture.

Ablation study

we verify the proposed innovation through ablation experiments on the Police Gesture Dataset. Specifically, an ablation experiment is conducted on each step of NTPGR, evaluating its significance in the overall algorithm by sequentially removing each step. The results are shown in Table 5. The findings reveal that when utilizing the MSNet network alone for gesture recognition, the accuracy only reaches 92.60%. After preprocessing the gesture dataset, the recognition accuracy with the MSNet network increases by 0.85%. This indicates that data preprocessing plays a vital role in gesture recognition tasks, as it effectively removes redundant information, enabling the model to learn key gesture features more accurately. Subsequently, when the EPFFNet network is integrated to extract spatial contextual features from gesture data (assuming the keypoint coordinate information had been modeled and fused), the accuracy of gesture recognition improves by 3.30% with the MSNet network. This highlights that the effective integration of spatial and temporal features in gesture data can significantly enhance the model's recognition performance. Finally, when all three components are combined to create the complete NTPGR, the recognition accuracy increases by an additional 4.96%. This finding indicates that gesture recognition depends not only on the temporal characteristics of dynamic gesture data but also on the effective utilization of spatial features. Additionally, enabling the network to focus on important information while eliminating redundant data will further enhance the model's performance in gesture recognition tasks.

To demonstrate the significance of each branch in the MSNet network designed in this study, each branch is sequentially removed to assess its impact on the performance of the gesture recognition task. The ablation

Method	Branch			Accuracy/%
	Convolution Network Branch	Sequence Network Branch	Hybrid Attention Branch	
SeriesNet				91.13
MSNet	✓			90.76
		✓		93.10
	✓	✓		96.40
	✓		✓	93.55
		✓	✓	95.60
	✓	✓	✓	**97.56**

Table 6. Ablation study were conducted on each branch of MSNet. The results are based on data preprocessing, gesture spatial feature extraction using the EPFFNet network, and feature modeling and fusion.

Method	Accuracy/%
NTPGR(-MECA)	94.35
+CA	94.67
+CABM	96.48
+Efficient Channel Attention(ECA)	93.50
+Efficient Coordinate Attention(ECA)	96.15
+MECA	**97.56**

Table 7. Ablation study of attention modules in Hybrid Attention Branch.

experiment is conducted after preprocessing the gesture data and using EPFFNet to extract gesture spatial features. The results are presented in Table 6. When utilizing only the convolution network branch, the model's accuracy decreased slightly. In contrast, using only the temporal network branch results in an accuracy increase of 1.97%. This indicates that the model's memory capability is essential for dynamic gesture recognition tasks. Ultimately, when the convolution network branch, the sequential network branch, and the hybrid attention branch are combined to form a complete MSNet network, the accuracy shows significant improvement compared to the baseline network. This suggests that the collaboration of each branch can effectively enhance the model's recognition performance.

Finally, to assess the impact of the MECA attention integrated into the hybrid attention branch on the traffic police gesture recognition task, we conduct ablation experiments on different attention mechanisms[43] using the Police Gesture Dataset. The experimental results are presented in Table 7. While maintaining the other innovative modules, the MECA attention is removed from the hybrid attention branch, resulting in a recognition accuracy of 94.35% for the modified network. Subsequently, when the hybrid attention branch incorporates CA attention, the improvement in accuracy is minimal. After adding CABM attention, the accuracy improves by 2.13%. The inclusion of efficient channel attention has a slight negative impact on the network's recognition performance. However, introducing efficient coordinate attention lead to an accuracy increase of 1.8%. When MECA attention is added, the network's recognition accuracy rise by 3.21%. This indicates that integrating MECA attention into the hybrid attention branch can effectively improve the network's gesture recognition capabilities.

Analysis of visualization

we utilize dynamic gesture data from the Police Gesture Dataset, focusing on complex scenes featuring passing pedestrians, trees, railings, and vehicles for visual recognition. The results are presented in Figure.8. The visualization indicates that the proposed algorithm achieves excellent recognition results across various scenarios. It not only accurately identifies the eight hand gestures of traffic police but also efficiently completes recognition despite external interference factors. This highlights that the NTPGR exhibits strong gesture recognition capabilities and real-time performance while maintaining a compact model size and low computational requirements, making it well-suited for practical deployment. Furthermore, we implement NTPGR for recognizing eight traffic police gestures, with the results illustrated in Figure.9. The findings indicate that NTPGR achieves a recognition accuracy exceeding 96% for these gestures, with minimal recognition errors for each gesture. This further demonstrates that NTPGR maintains high precision across different types of traffic police gesture recognition, confirming the algorithm's effectiveness in this domain.

Model failure case analysis

Although NTPGR performs well in traffic police gesture recognition tasks, misclassifications still occur in certain scenarios. To better understand the limitations of NTPGR, we analyze the misclassification cases and identify the following primary types of errors:

- **Confusion due to gesture similarity:** Some traffic police gestures are visually similar, especially when there are only slight angle changes or finger position differences. This can cause confusion in the model. For exam-



Fig. 8. Qualitative results for NTPGR in the Police Gesture Dataset. Illustrating Visual representation of NTPGR's performance in complex environments influenced by special factors.

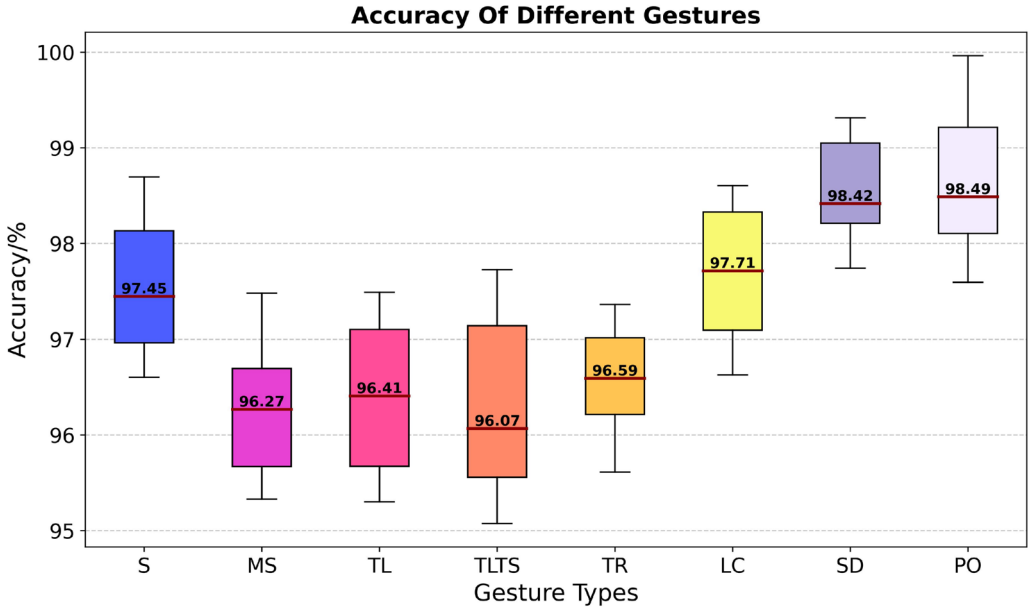
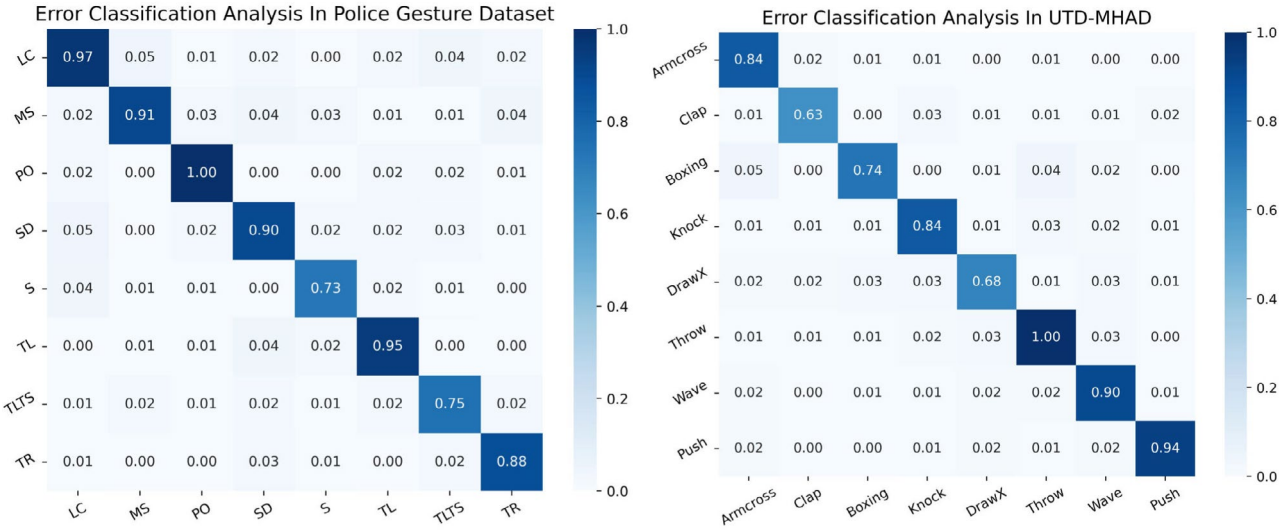


Fig. 9. Box plot illustrating the accuracy of NTPGR in recognizing different gestures in the Police Gesture Dataset.

ple, in the Police Gesture Dataset, "Move Straight" and "Turn Left" gestures share similar structures, and in certain shooting angles, misclassifications may occur.

- **Background interference:** When the background environment is complex or lighting conditions fluctuate, the model's recognition ability can be compromised. For instance, when the background color is similar to the gesture or the texture features are complex, the model may struggle to accurately segment the gesture area. Furthermore, dynamic background interference, such as moving pedestrians or vehicles, can disrupt the extraction of gesture features, leading to misclassification.
- **Lighting and perspective changes:** Extreme lighting conditions and non-standard angles can adversely affect recognition results. In low-light environments, gesture edges may become blurred, leading to unstable feature extraction. In bright lighting, parts of the gesture may become overexposed, causing the loss of critical infor-



(a) Misclassifications in Police Gesture Dataset **(b)** Misclassifications in UTD-MHAD dataset

Fig. 10. A error classification analysis of NTPGR based on different gestures. Result measured in the Police Gesture Dataset and UTD-MHAD dataset.

mation. Additionally, gestures may be obscured in overhead or low-angle views, making accurate recognition more difficult.

- **Instability in recognizing minority gesture classes:** Due to an insufficient number of data samples for some gesture categories, the model's generalization ability in these categories is weaker, leading to unstable classification. For example, in the Police Gesture Dataset, "Turn Left" and "Turn Left to Stay" gestures have imbalanced data distributions, causing lower distinguishability between these two gestures and increasing the likelihood of misclassification.

To further quantify and analyze misclassifications, we plot the confusion matrices for NTPGR on both the Police Gesture Dataset and the UTD-MHAD Dataset, as shown in Figure.10. From these matrices, we observe that there is significant confusion between certain gesture categories. For instance, in both the Police Gesture Dataset and the UTD-MHAD Dataset, the probability of misclassifying "Move Straight" as "Slow down" and "Armcross" as "Boxing" reached 5%. This indicates that the feature distinguishability between some gestures in the datasets still has room for improvement.To address these issues, we plan to incorporate contrastive learning in future research to enhance the model's ability to differentiate between similar gestures. Additionally, we will adopt data balancing strategies to reduce recognition bias in minority gesture classes. Furthermore, optimizing the feature extraction module to improve the model's robustness in complex environments will also be a key focus in future optimizations.

Conclusions

This paper introduces a novel traffic police gesture recognizer (NTPGR), designed to address the shortcomings of existing gesture recognition technologies by providing a lightweight and efficient solution. NTPGR integrates spatial-temporal features through four key steps for efficient recognition of traffic police gestures: gesture data preprocessing, gesture spatial feature extraction, spatial feature modeling and fusion, and gesture time sequence feature extraction. Through the organic combination of these steps, NTPGR demonstrates exceptional performance, even in complex scenarios such as occlusion and dynamic backgrounds. In terms of network architecture, NTPGR adopts a hybrid structure composed of the Efficient Progressive Feature Fusion Network(EPFFNet) and the Multi-Branch Gesture Recognition Network (MSNet). EPFFNet efficiently extracts the skeletal data of traffic police, with its innovative progressive feature fusion module ensuring the capture of high-resolution spatial details under low computational load. Additionally, we improve the baseline SeriesNet by introducing a convolution branch specifically for traffic police gesture recognition, which precisely captures the spatial context of human posture, further optimizing recognition performance. To enhance focus on critical frames, we design a hybrid attention branch consisting of temporal and channel attention modules, enabling the model to assign adaptive attention weights to different key frames and better capture important gesture features. MSNet, with its multiple branches, further extracts and fuses temporal features, ensuring multi-layered, multi-angle feature fusion that significantly improves overall recognition accuracy. Experimental results show that NTPGR achieves recognition accuracies of 97.56% and 96.76% on the Police Gesture Dataset and the UTD-MHAD Dataset, respectively, with average response times of just 0.76 s and 0.74 s, demonstrating its competitive edge in both performance and real-time capability. These results not only provide a solid technical foundation for gesture recognition in autonomous driving systems but also contribute to enhancing the intelligence and responsiveness of autonomous vehicles.

However, despite the above achievements, there are still challenges to be addressed in practical deployment. Issues such as misclassification, difficulty distinguishing similar gestures, and the potential impact of data distribution imbalance remain pressing concerns. Additionally, autonomous driving systems must take ethical and legal considerations into account to ensure their safety and fairness in intelligent traffic management. Therefore, future work will focus on the following improvements:

- Further optimizing the model structure to reduce misclassification rates and enhance adaptability in complex environments.
- Introducing uncertainty estimation and anomaly detection mechanisms to mitigate the negative impact of misidentifications on system performance.
- Investigating the ethical and legal aspects of the system in real-world applications to ensure compliance with safety, legality, and fairness requirements in intelligent traffic management.

Through these improvements, we aim for NTPGR to be applied in a broader range of real-world scenarios, providing more solid and reliable technical support for real-time, accurate gesture recognition in future autonomous driving systems.

Data availability

The datasets generated during and/or analyzed during the current study are not publicly available due to [Data and processing information related to product development] but are available from the corresponding author on reasonable request.

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Author contributions

J.X. conceives the innovation(s), conducts the experiment(s), writes original manuscript and processes data, H.L. and J.Z. provides financial support and analyzes the results, H.L. and J.Z. reviewes the manuscript, all authors reviewes the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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